

What is HTTP

Protocol Rules - (Pre Installed Rules in OS)

With Some RULES Client - Server Communicate

HTTP Versions

HTTP (1.0, 1.1, 2.0, 3.0)
(Slow, Fast, Faster, Fastest)

Status Codes

(200, 301, 404, 500)
(Success, Redirect, Not Found, Server Error)

Why HTTPS -
Hyper Text Transfer Protocol Secured

In HTTP (Man in the Middle can See Over Communication Data)

In HTTPS (Encryption is Used For Secure Communication)-Even
If Hacker See it He cant Understand it Bcz of Encryption

HTTPS adds Security, Encrypts Data, Protects Passwords and
Payments - Using Some Methods (SSL/TLS Encryption,
Handshake = Shared Key, Safe Tunnel)

HTTPS Uses SSL/TLS. Browser and Server First Do a Handshake
to agree on a secret Key. After That, all data Trough an
Encrypted Tunnel - only Browser and Server can Unlock it. Thats
why Bank and Payments sites always use HTTPS.

Proxy vs Reverse Proxy

A Proxy Server and a Reverse Proxy Server both act as intermediaries, but they serve opposite sides of a client–server communication.

1. Proxy (Forward Proxy)

Definition

**A proxy sits between the client and the internet.
It represents the client when making requests to servers.**

How it works

Client → Proxy → Server

Main Purposes

- **Client anonymity (hides client IP)**
- **Access control (restrict websites)**
- **Caching to speed up repeated requests**
- **Security (filter malicious websites)**

Used By

- **Individuals**
- **Organizations restricting employee internet usage**
- **Schools, offices**

Example

A company uses a proxy to allow only approved sites for employees.

2. Reverse Proxy

Definition

**A reverse proxy sits between the server and the internet.
It represents the server, handling client requests on behalf of**

backend servers.

How it works

Client → Reverse Proxy → Backend Server(s)

Main Purposes

- **Load balancing (distribute traffic across multiple servers)**
- **Security (hide server IP, protect from DDoS)**
- **Caching (store static responses to improve speed)**
- **SSL termination (handle HTTPS encryption)**

Used By

- **Large websites**
- **Web applications behind load balancers**
- **CDNs (Cloudflare, Akamai)**

Example

Cloudflare acts as a reverse proxy to protect and accelerate websites.

1. Proxy (Forward Proxy) — Like a Personal Assistant

You (the client) don't go directly to the hotel's shops or services. Instead, you tell your personal assistant to fetch things for you.

How it works

- **You → Personal Assistant (Proxy) → Hotel shops**
- **The shop doesn't know *you*; it only sees the assistant.**

Why use it?

- **You want privacy (hide your identity)**
- **You want restricted access (assistant blocks certain shops)**
- **Assistant can remember frequently bought items (caching)**

Example

Students in a college access the internet through a proxy that blocks social media.

2. Reverse Proxy — Like a Hotel Reception Desk

Now think of a big hotel with many internal departments. Guests (clients) never interact directly with those departments.

They go to the reception desk (reverse proxy). The receptionist decides which department should handle the request.

How it works

- **Guest → Reception (Reverse Proxy) → Internal departments**
- **Reception hides the internal layout of the hotel.**

Why use it?

- **Distribute guests among departments (load balancing)**
- **Protect departments from direct interaction (security)**
- **Reception can handle common questions directly (caching)**
- **Reception manages all key verifications (SSL termination)**

What is a VPN?

A VPN (Virtual Private Network) is a secure tunnel that encrypts your internet traffic and hides your identity by routing your connection through a remote server.

Main functions

- **Hides your IP address**
- **Encrypts your data**
- **Protects you on public Wi-Fi**
- **Lets you access blocked/restricted content**

★ Analogy: VPN as a Secret Tunnel Through a City

Imagine the internet is a big city full of roads.

Normally, when you walk on the streets:

- People can see you
- They know where you're going
- They can track your path

This is like browsing the internet without a VPN.

🔒 With a VPN – You Use a Secret Tunnel

Now imagine there is a private underground tunnel:

- Nobody in the city can see you enter or where you're going
- Your path is hidden
- You pop out of the tunnel from another location (maybe even another country!)

This is exactly what a VPN does.

✅ How Proxy Works at Browser Level

A proxy server is typically configured inside a single application, usually your browser.

Meaning

Only the browser's traffic passes through the proxy.

Other apps (WhatsApp, YouTube app, system updates, games, etc.) do NOT use the proxy unless you configure them separately.

Why?

Because proxies operate at the application layer (Layer 7) of the

OSI model.

★ Analogy (Browser-Level Proxy)

Proxy = Shop Assistant for Only One Shop

Imagine you go to a shopping mall and hire an assistant only for one shop—for example, the electronics store.

- Whenever you enter that store, the assistant helps you.
- But when you go to other stores, the assistant does NOT follow you.

This is exactly a proxy:

➡ It works only for the app where you configure it.

✅ How VPN Works at Device Level

A VPN intercepts all network traffic at the operating system level, not just for a single app.

Meaning

All apps on your device use the VPN automatically:

- Browsers
- Games
- Banking apps
- System updates
- Background services

Everything goes through the VPN's encrypted tunnel.

Why?

VPN operates at the network layer (Layer 3) of the OSI model, below applications.

★ Analogy (Device-Level VPN)

VPN = Tunnel for the Entire House

Imagine your house has a private underground tunnel connected to the outside world.

- **Everyone in the house (you, your family, guests) has to use this tunnel.**
- **Nobody can go outside through any other route.**

This is exactly how a VPN works:

 **Every app on your device uses the VPN tunnel automatically.**